# Elbaraa Refaie

Senior Automation Engineer | Packaging & EOL Systems

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## Summary

***Senior Automation Engineer | Packaging & End-of-Line Systems | Schneider PacDrive Specialist***

*Results-driven Automation Engineer with 3+ years of end-to-end ownership of EOL packaging machinery — from PLC architecture and servo motion programming through FAT, international commissioning, after-sales support, and customer-driven upgrades. Designed, commissioned, and actively supports 100+ machines for multinational FMCG manufacturers across 6 countries, maintaining 100–150 active support connections annually. Deep expertise in Schneider Electric PacDrive 3 ecosystem — LMC series controllers, MC12 MultiCarrier linear motor systems, multi-axis servo coordination, and reusable Function Block development including checkweigher integration. Architected and deployed an embedded two-layer Production Intelligence Platform (PacDrive 3 ST logic + Jmobile IPC reporting) running as standard across the Vervo machine portfolio — delivering OEE shift reporting with scheduled PDF export, machine state tracking with CSV diagnostics log, and a 24-hour production speed historian at 1-second resolution. Equivalent functionality to a dedicated MES and SCADA historian installation — at zero external software cost. Delivered flagship projects including a 550 PPM tray lid line (16+ servo axes, MC12, Schmalz vacuum, 12 glue guns) and a proprietary Smart Infeed System merging 5 lines at 400 PPM total throughput. Published researcher in machine vision and AI; B.Sc. Electrical-Mechatronics Engineering, CGPA 3.94/4.0.*

## Experience

### VERVO FOR ADVANCED PACKAGING, Istanbul, Turkey

(Machine Builder — End-of-Line Packaging Systems)

AUTOMATION ENGINEER *NOV 2022 – Present*

*Sole automation engineer responsible for the full programming lifecycle of EOL packaging machinery — from software architecture and PLC development through FAT, site commissioning, and post-sales support — serving major FMCG manufacturers across 6 countries.*

- *Programmed and commissioned **100+ EOL packaging machines** including box erectors, case packers, and tray lid systems for clients including Ülker, AdkuTurk (Indomie TR), Tadım, Torku, Tadım Germany, Meray Germany, Indomie Saudi Arabia & Serbia, Hayat Kimya, Eruslu, BIM, Mutlu Un, İpek Pamuk* and others
- *Designed and commissioned a **550 PPM high-speed** tray lid packaging line at Ülker Gebze — coordinating 16+ servo axes, 7 conveyors, **MC12 MultiCarrier**, Schmalz vacuum gripper, 12 hot-melt glue guns, and an automated reject module for length control and overlap detection, all on Schneider PacDrive 3 with Exor Jmobile HMI*
- Developed a proprietary Smart Infeed System (Akıllı Ürün Taşıma Sistemi) capable of dynamically merging 2–5 independent upstream production lines (each at 80 PPM) into a single synchronized 400 PPM feed for one case packing machine — eliminating the need for multiple case packers and significantly reducing client capital and floor space costs
- *Developed all machine software on **Schneider PacDrive 3 / LMC series** platform using **IEC 61131-3 Structured Text**, covering motion control, safety logic, recipe management, and fault diagnostics*
- *Designed and deployed **operator HMIs** with multilingual support,* reducing operator error rate by an estimated 30% and cutting average fault-resolution time through structured alarm management and multilingual interface design
- *Established **standardized reusable code libraries** for common machine functions,* reducing average PLC programming time per new machine by approximately 25%, enabling faster project delivery across repeat machine types
- *Led FAT and SAT for 100+ machines with full documentation delivery — including functional specs, parameter backups, and test protocols — across 6 countries: Turkey, Germany, Saudi Arabia, Serbia, Egypt, and Malaysia*
- Designed and deployed an embedded two-layer Production Intelligence Platform across the Vervo machine portfolio — PacDrive 3 Structured Text FB library handling all KPI calculation and state logic, with Exor Jmobile IPC managing storage, reporting, and visualization — comprising:
    1. OEE Reporting: shift-configurable tracking of products, rejections, boxes, stops, homing events, and stop time — with on-demand and scheduled PDF export from HMI
    2. Machine State Tracker: continuous logging of all state transitions with time-in-state, product/box counters, and CSV export for remote fault diagnosis
    3. Production Speed Historian: 24-hour rolling trend at 1-second resolution — functioning as an embedded SCADA historian natively on the Jmobile IPC
- *Delivered **operator and maintenance training** to client engineering teams post-commissioning,* reducing first-month downtime incidents through structured handover training and commissioning documentation delivered to client maintenance teams.
- *Developed reusable Function Block libraries for **checkweigher system integration** (GMT-P1 loadcell) — enabling standardized weight-based reject logic across multiple machine types and reducing integration time for new checkweigher connections*
- *Managed **full after-sales service lifecycle for 100+ installed machines** — including fault diagnosis, software corrections, parameter recovery, and emergency remote intervention*
- *Led **machine upgrade** projects based on customer requirements —adding new product recipes, expanding format ranges, and integrating new peripheral modules into existing PacDrive 3 software architecture without production line interruption*
- *Trained internal engineering team members, our operators and aftersales members a on PacDrive 3 platform operation, HMI usage, fault troubleshooting, and preventive maintenance procedures*
- *Maintained an average of **100–150 active machine support connections per year** across installed base — providing consistent technical coverage across Turkey, Germany, Arabia, Serbia and Egypt*

***Key Technologies:** Schneider PacDrive 3, LMC106/212,216, LMC 802/902 Pro, MC12 MultiCarrier, EcoStruxure Machine Expert, Structured Text (ST), Servo Motion Control, HMI (Exor Jmobile)*

### STUDYFANS COMPANY, Istanbul, Turkey

*Business Data Analyst* *JUN 2022 – OCT 2022*

- *Automated repetitive reporting workflows using Python, reducing manual data processing time by an estimated 60%*
- *Built Tableau dashboards tracking KPIs for department head visibility*
- *Developed standardized documentation and workflow procedures for the department*

### TOP GLOVE CORPORATION, Malaysia

Automation & Digitalization Intern *JULY 2021 – OCT 2021*

- *Implemented real-time sensor dashboards for bulk tank monitoring, supporting operational decision-making*
- *Built inventory management applications using Microsoft PowerApps*
- *Collaborated with cross-functional automation teams on live production systems*

### CENTER OF ROBOTICS AND ARTIFICIAL INTELLIGENCE (CAIRO UTM)

RESEARCH ASSISTANT MARCH 2021 – JAN 2023

- *Co-authored 2 peer-reviewed publications in machine vision and AI-based traffic analysis (IOP Publishing; Springer)*
- *Developed computer vision systems for vehicle classification and counting using YOLO and DeepSORT. Result*
- *Engineered a facial gesture-based computer control system for amputee accessibility. Result*
- *Prof. Ahmed Athif Recommendation -Lab Director - Link.*

## Projects

1. **KEY PROJECT: HIGH-SPEED TRAY LID PACKAGING LINE — 550 PPM**

- **Platform :** *Schneider PacDrive 3 | LMC Series | MC12 MultiCarrier System | EcoStruxure Machine Expert | Exor Jmobile HMI*
- **Scope :** *Full PLC software development, servo motion tuning, HMI design, FAT, and live commissioning*
- **Machine Architecture:**
  - *16+ servo motors — fully synchronized multi-axis motion control, 7 conveyors — product transport, spacing, and accumulation, MC12 MultiCarrier linear motor system — high-speed carton transport*
  - *Schmalz vacuum gripper system — product pick and place, Carton picker and transfer module, Folding module*
  - *Multibelt system (Schneider) — synchronized belt drive coordination*
  - *Reject module — automatic ejection of defective products based on length control and product overlap detection*
- **Control Challenges Solved:**
  - *Real-time synchronization of 16+ servo axes with zero product collision tolerance*
  - *Dynamic reject logic combining length measurement and detection sensors — preventing defective product reaching the case packer*
  - *Coordinated glue gun firing timing across 12 guns with carton position on MC12 carriers*
  - *Recipe management system covering multiple product formats and carton sizes with single-operator changeover*
  - *Multilingual HMI (Exor Jmobile) with structured alarm management, production counters, and fault diagnostics*
  - *Full FAT at Vervo facility + live SAT commissioning at Ülker Gebze production floor*
- ***Result:***
  - *Machine operating in live production at Ülker Gebze with speed up to 500 ppm among the highest-speed tray lid lines in Turkish FMCG packaging. Commissioned and handed over with full operator and maintenance documentation. [Watch here]*

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2. **SMART INFEED SYSTEM**

- **Platform :** *Schneider PacDrive 3 | LMC Series | EcoStruxure Machine Expert | Servo Motion Control | Multi-Axis Synchronization*
- **Challenge :** *Customer production lines running 5 parallel upstream conveyors at 80 PPM each — previously requiring 5 case packers.*
- **Solution :** *Designed and programmed a proprietary smart collation and infeed that dynamically merges 2 to 5 independent product lanes into a single synchronized product stream, feeding one high-speed case packing machine.*
- **Capability:**
  - *Total throughput: up to 400 PPM (5 lanes × 80 PPM)*
  - *Real-time speed synchronization across all active upstream lines*
  - *product spacing and gap correction before case packer infeed*
  - *Flexible enough to handle partial line shutdowns without stopping packer*
- **Result:**
  - *Enabled customers to replace multiple case packers with a single high-efficiency machine — reducing capital cost, floor space, and maintenance overhead significantly.*

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3. **EMBEDDED PRODUCTION INTELLIGENCE PLATFORM**

- **Platform :** *PacDrive 3 Structured Text (logic layer) + Exor Jmobile IPC (storage, reporting, visualization)*

*A two-layer embedded platform deployed as standard across the Vervo machine portfolio — delivering MES/SCADA-grade production visibility at zero external software cost:*

- *OEE Reporting — shift-configurable KPI tracking (products processed/rejected, boxes produced, machine stops, homing events, total stop time) with on-demand and scheduled PDF report generation from HMI*
- *Machine State Tracker — continuous logging of all state transitions (Auto/Stop/Reference/Fault/E-Stop) capturing time-in-state and production counters at each event, exported as CSV for remote fault diagnosis*
- *Production Speed Historian — 24-hour rolling speed trend at 1-second resolution (86,400 points/day) visualized natively on Jmobile — equivalent to a dedicated SCADA historian node per machine*

- **Result:** *Clients receive embedded production intelligence with zero MES licensing cost. CSV exports enable remote fault diagnosis without site visits, reducing downtime and travel overhead across 6 countries.*

## Skills

AUTOMATION & CONTROL

- *Schneider Electric PacDrive 3 | EcoStruxure Machine Expert | LMC Series (106/212/216, 802/902 Pro)*
- *MC12 MultiCarrier | Servo Motion Control | Multi-Axis Coordination*
- *IEC 61131-3 Structured Text (ST) | Function Block (FB)*
- *HMI Development — Exor Jmobile | Vijeo Designer*
- *FAT / SAT | Machine Commissioning | End-of-Line (EOL) Automation*
- *OEE Monitoring*
- *CANopen | EtherCAT | Modbus TCP*
- *SCADA concepts | OPC-UA*
- *Fieldbus communication*
- *Change management / format changeover*

PROGRAMMING & SOFTWARE

- *Python | Embedded C/C++ | EcoStruxure Machine Expert*
- *OpenCV | YOLO | DeepSORT | TensorFlow (basic)*
- *Tableau | SolidWorks*
- *PCB Design: Altium Designer, Proteus | Circuit Simulation: Multisim*

## Publications

1. Loon, Lee Pin, Elbaraa Refaie, and Ahmad Athif Mohd Faudzi. "Data analytics for traffic flow prediction in Custom using Long Short-Term Memory (LSTM) networks." *Journal of Physics: Conference Series*. Vol. 2107. No. 1. IOP Publishing, 2021. Retrieved from: https://doi.org/10.1088/1742-6596/2107/1/012006

2. Refaie, E., & Faudzi, A. A. M. (2022). Machine Vision-Based System for Vehicle Classification and Counting Using YOLO. In *Computational Intelligence in Machine Learning* (pp. 391-398). Springer, Singapore. Retrieved from: https://doi.org/10.1007/978-981-16-8484-5_37

## Education

### B.Sc. University of Technology Malaysia (UTM)

B.S. IN ELECTRICAL ENGINEERING — *Sep 2017 – Feb 2022*

- *CGPA 3.94/4.0*
- *Final Year Project Title: Integrated traffic system for vehicles classification and counting and traffic prediction using DeepSORT and LSTM algorithms.*
- *Final Year Project Thesis. Link.*
- *Final Year Project Grade: A+*

### B.Sc. Zagazig University in Egypt

B.S. IN MECHANICAL ENGINEERING — *Sep 2014 – Apr 2016*

- *Completed 3 semesters with 89% before pursuing Electrical-Mechatronics Engineering at UTM, Malaysia*

## Honours & Awards

- *Robotics Competition Awards: Champion & 2nd Runner-Up, U-ROCK UNIMAP (2019); Best Engineering Award & 1st Runner-Up, Malaysia Robocon (2019); 1st Runner-Up, University AI Hackathon (2020)*

## Languages

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| Arabic | *Native speaker (Mother Tongue)* |
| English | *Professional verbal and written communication skills*<br>*IELTS Test Score: 6.5* |
| Turkish | *Above Average* |

## References

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| Baris Menderes | *Vervo Development Manager* | *baris.menderes@vervo.com.tr* | |
| Prof. Dr. Zaharuddin Mohamed | *Head of Advanced Control Research Group, School of Electrical Engineering*<br>*My Academic Advisor* | Contact details available upon request | Google Scholar |
| Prof. Ir. Ts. Dr. Ahmad 'Athif Bin Mohd Faudzi | Director of Centre for Artificial Intelligence and Robotics (CAIRO UTM)<br>*My Final Year Project Supervisor* | Contact details available upon request | Google Scholar |